

Econ 2203 | International Trade and Policy in Agriculture

Department of Development Economics

What this course is about

- The economics of trade between nations — with a focus on **agricultural commodities**
- Why countries trade, who gains, who loses, and what policy should do about it
- India's position in global agricultural trade and key policy instruments

Learning objectives

- Master classical and modern theories of trade
- Analyse India's agricultural trade flows and policy environment
- Evaluate trade policy tools: tariffs, quotas, subsidies, SPS measures
- Apply WTO frameworks to real agricultural trade disputes

Assessment: Mid-semester exam (30%) · End-semester exam (40%) · Term paper (20%) · Participation (10%)

Course Roadmap: Lectures 1–9

Lec	Topic	Date
1	Introduction to International Trade	Apr 28
2	Features & Internal vs International Trade	May 5
3	Theories I — Mercantilism & Absolute Advantage	May 12
4	Theories II — Comparative Advantage	May 19
5	Theories III — Factor Endowments (H-O Model)	May 26
6	New Trade Theory & Intra-Industry Trade	Jun 2
7	Terms of Trade & Trade Gains	Jun 9
8	Trade Policy: Tariffs	Jun 16
9	Trade Policy: Non-Tariff Barriers	Jun 23

Course Roadmap: Lectures 10–18

Lec	Topic	Date
10	Balance of Payments & Exchange Rates	Jun 30
11	WTO Framework & Agreements	Jul 7
12	Agreement on Agriculture (AoA)	Jul 14
13	SPS & TBT Measures in Agriculture	Jul 21
14	India's Foreign Trade Policy	Jul 28
15	Regional Trade Agreements — ASEAN, SAFTA	Aug 4
16	Agricultural Price Policy & Trade	Aug 11
17	Food Security and Trade Nexus	Aug 18
18	Contemporary Issues & Revision	Aug 25

What is International Trade?

Definition

*International trade is the **cross-border exchange** of goods, services, and capital between residents of different countries.*

Three dimensions of trade

- **Merchandise trade** — physical goods: rice, wheat, machinery, petroleum
- **Services trade** — IT services, tourism, financial services, education
- **Capital flows** — foreign direct investment, portfolio investment, remittances

Why it matters for agricultural economists

- India earns ~₹3.6 lakh crore from agricultural exports annually
- Global price signals directly affect domestic farm prices
- WTO rules constrain domestic agricultural support policies

Core insight: Trade is fundamentally about countries doing what they are relatively better at, and exchanging the surplus — creating gains for all participants.

Why Do Nations Trade?

Supply-side reasons

- **Resource endowments** — India has fertile Indo-Gangetic plains; Brazil has Amazon basin → different agricultural outputs
- **Technology differences** — Netherlands uses precision horticulture; India relies on labour-intensive methods
- **Specialization gains** — Focusing on fewer products reduces unit cost
- **Economies of scale** — Large production runs lower average cost (e.g., Thai rice mills)

Demand-side reasons

- **Product variety** — Consumers want Darjeeling tea *and* Colombian coffee
- **Seasonal complementarity** — Southern hemisphere harvests when Northern hemisphere is off-season

Global merchandise trade (2023)

- Total value: ~\$25 trillion (WTO, 2024)
- Services trade: ~\$7 trillion
- Combined: ~\$32 trillion — roughly 30% of world GDP

Rank	Country/Bloc	Share	Year	Value (USD trillion)
1	China	14.2%	1950	0.06
2	United States	11.3%	1990	3.5
3	European Union	14.8%	2010	15.3
—	—	—	2023	25.0

Factoid: Global merchandise trade contracted only twice since 1950 — during the 2008–09 financial crisis and the 2020 COVID shock.

India's trade position (FY2024)

Indicator	Value
Merchandise exports	\$437 billion
Merchandise imports	\$678 billion
Merchandise trade deficit	-\$241 billion
Services exports	~\$340 billion
Services surplus	+\$160 billion
Total merchandise trade	\$1.18 trillion

Global rank: 18th in merchandise exports (WTO 2023)

India's Trade: The Big Picture

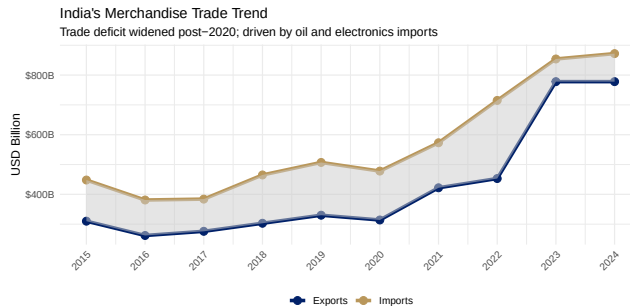


Figure 1: India's Merchandise Trade (FY2015–FY2024, USD billion) Source: DGCI&S / Ministry of Commerce and Industry (GoI).

India's Agricultural Trade: FY2024

Top agricultural exports and imports

Commodity (Exports)	Value (USD bn)	Commodity (Imports)	Value (USD bn)
Rice (Basmati + Non-Basmati)	10.4	Edible oils (palm, soy)	12–14
Marine products	7.6	Pulses	3.0
Spices	3.7	Cashew (raw)	1.8
Buffalo meat	3.5	Fresh fruits	1.5
Sugar	2.8	Spices (select)	0.8
Total agri exports	~43.7	Total agri imports	~28.2

Agricultural trade surplus: ~\$15.5 billion

Source: APEDA / DGCI&S, 2024

India's Agricultural Trade Profile

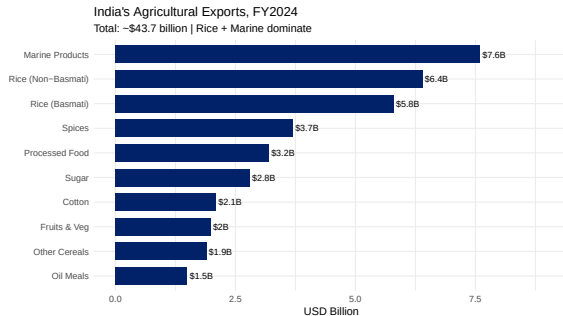


Figure 2: India's Agricultural Exports by Commodity, FY2024 (USD billion) Source: APEDA Annual Report 2023–24.

Export side

- Agriculture: ~11% of total merchandise exports (\$43.7B of \$437B)
- Engineering goods: ~26% | Petroleum products: ~17% | Gems & jewellery: ~10%
- Agriculture is **India's second-largest net foreign exchange earner** after services
- **India's strengths:** rice, spices, marine products, cotton — all labour-intensive, climate-favoured commodities

Import side

- Edible oils: India imports **60–65%** of its edible oil requirement (palm, soybean, sunflower)
- Pulses: Canada, Australia, Myanmar — ~25% of domestic requirement imported

Vulnerability: India's dependence on imported edible oils (~\$13B/yr) exposes food inflation to global price shocks, as seen in 2021–22 when palm oil prices spiked 50%.

India's Position in Global Agricultural Trade

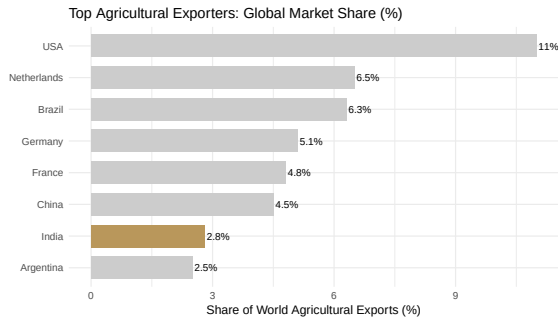


Figure 3: Major Agricultural Exporters: Global Market Share (2023) Source: WTO, World Agricultural Trade Statistics 2024.

Classical trade theory

- **Comparative advantage** — trade based on *relative*, not absolute, efficiency (Ricardo)
- **Factor endowments** — land, labour, capital abundance shapes what countries export (Heckscher-Ohlin)
- **Terms of trade** — ratio of export prices to import prices; determines gains from trade
- **Intra-industry trade** — two-way trade in similar products (India exports basmati, imports parboiled rice)

Trade policy instruments

- **Tariff** — tax on imports; raises domestic price above world price
- **Non-Tariff Barrier (NTB)** — quotas, SPS standards, licensing
- **Balance of Payments (BoP)** — accounting of all international transactions
- **Exchange rate** — price of one currency in terms of another; stronger rupee makes exports costlier

Policy relevance for India

- **Foreign Trade Policy 2023–28** targets \$2 trillion in merchandise + services exports by 2030
- **WTO commitments** constrain India's ability to protect domestic agriculture (AMS limits; tariff bindings)
- **Farmer income** — ~50% of India's workforce in agriculture; export markets expand income
- **Food security** — import policy on pulses and oilseeds directly affects 1.4 billion people

What Makes Agricultural Trade Special?

Biological & physical features

- **Perishability** — fresh produce, milk, meat have short shelf-lives → need cold chains, quick clearance
- **Seasonality** — harvests are discrete events; supply is lumpy
- **Bulkiness** — grain, sugar, cotton: high weight-to-value ratio → high freight costs
- **Sanitary risks** — food-borne pathogens cross borders with commodities → SPS measures justified

Economic & political features

- **Food security concern** — governments intervene to ensure domestic supply; export bans, import duty waivers are common
- **WTO special treatment** — Agreement on Agriculture has Special and Differential Treatment for developing countries

India's Export Destinations (FY2024)

Top merchandise export destinations

Rank	Country	Share of exports	Key agri exports
1	United States	~18%	Marine, spices
2	UAE	~6.5%	Rice, meat, fruits
3	Netherlands	~4.2%	Spices, cotton
4	China	~3.8%	Cotton, castor oil
5	Bangladesh	~3.5%	Rice, onion, sugar

Top merchandise import sources

Rank	Country	Share of imports	Key agri imports
1	China	~15%	Electronics, chemicals
2	UAE	~7.5%	Crude oil, gold
3	USA	~6.2%	Pulses, capital goods
4	Russia	~5.8%	Crude oil, fertilisers
5	Saudi Arabia	~5.4%	Crude oil, edible oils

Composition of India's Merchandise Exports

Major export categories (FY2024, approx.)

Category	Value (USD bn)	Share
Engineering goods	112	25.6%
Petroleum products	78	17.8%
Gems & jewellery	44	10.1%
Chemicals	36	8.2%
Agriculture & allied	44	10.1%
Textiles & apparel	35	8.0%
Electronic goods	29	6.6%

- Engineering goods grew from 12% (2000) to 26 (2024)
- Agriculture has held a **stable ~10–12%** share for two decades

Foreign Trade Policy (FTP) 2023–28: Key pillars

1. **Export target:** \$2 trillion merchandise + services exports by 2030
2. **Districts as Export Hubs (DEH):** Identify 1 product per district with export potential; provide targeted support
3. **ECGC coverage:** Expanded Export Credit Guarantee Corporation cover to de-risk exporters, especially MSMEs
4. **Remission of Duties and Taxes on Exported Products (RoDTEP):** Replace old MEIS; neutralise embedded taxes
5. **E-commerce exports:** Dedicated policy for cross-border e-commerce; target \$200–300B by 2030

Agricultural trade-specific provisions

- Agri-specific chapters supporting APEDA-registered exporters
- GI tag promotion (Darjeeling tea, Alphonso mangoes, Basmati rice)
- Simplification of phytosanitary certification processes

FTP 2023-28 is notably continuity-oriented — it builds on previous policy architecture rather than a radical overhaul, reflecting India's cautious approach to trade liberalisation.

PPF — conceptual preview

Without trade (autarky)

- A country produces *and* consumes on its PPF
- The slope of the PPF = the domestic relative price
- Example: India producing only rice and wheat — the trade-off is fixed by domestic technology

A Preview: The Basic Trade Model (II)

With trade (open economy)

- World price ratio differs from domestic price ratio
- Country specialises in the good where it has comparative advantage
- Consumption point can now lie **outside** the PPF — *both* countries gain

Trade flows

- **Exports** — goods/services sold to foreign residents; flow of goods out, money in
- **Imports** — goods/services bought from foreign residents; flow of goods in, money out
- **Trade balance** = Exports - Imports (positive → surplus; negative → deficit)
- **Visible trade** — physical merchandise (rice, cars, oil)
- **Invisible trade** — services, remittances, investment income

Balance of Payments

- **Current account** — trade in goods + services + remittances + investment income
- **Capital account** — financial flows: FDI, portfolio investment, loans
- **BoP identity:** Current account + Capital account = 0 (approximately)

Exchange rate

- **Nominal exchange rate** — ₹/USD; e.g., ₹83 = \$1
- **Real exchange rate** — adjusts for price level differences; determines *actual* competitiveness

Four reasons international trade is essential for India

1. **Growth engine:** Trade has been central to every development success story — Japan, South Korea, China, and now India's IT sector
2. **Consumer welfare:** Trade gives 1.4 billion Indians access to goods they cannot produce cheaply at home
3. **Farmer incomes:** Export markets for rice, spices, and marine products support millions of rural households
4. **Policy discipline:** WTO commitments create external pressure for domestic policy reform

What we covered today

- International trade = cross-border exchange of goods, services, and capital
- Global merchandise trade: **~\$25 trillion (2023)**; services: **~\$7 trillion**
- India ranks **18th** in merchandise exports; total trade **~\$1.18 trillion (FY2024)**
- Agricultural exports: **\$43.7 billion (FY2024)** — rice, marine, spices, cotton dominate
- India has a **structural agricultural trade surplus** (~\$15.5B) that helps offset the merchandise deficit

Questions to think about before next lecture:

1. Why does India ban rice exports during domestic inflation? Who benefits, who loses?
2. If India has absolute advantage in producing rice, why does Bangladesh also export rice?

Features of International Trade (*May 5, 2026*)

We will cover:

- **Salient features** of international trade that distinguish it from domestic trade
- **Immobility of factors** — why labour and capital don't flow freely across borders
- **Currency and exchange risk** — how rupee movements affect export competitiveness
- **Internal vs international trade** — a systematic comparison
- **The gravity model** — intuition and India examples

Further Reading

- *International Economics* — Salvatore (Ch. 1)
- *International Economics* — Appleyard & Field (Ch. 1)
- RBI/DGCI&S/APEDA databases for latest data

Key Data Sources

- DGCI&S: India's merchandise trade
- RBI: Balance of payments data
- APEDA: Agricultural export statistics
- WTO: Tariff and trade databases